

THE INSTRUCTION MANUAL
CosmicWatch
The Desktop Muon Detector (v3C)

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1 DOCUMENT OVERVIEW

This manual provides instructions for the setup, operation, and basic use of the CosmicWatch v3C particle detector. All material can be found in the GitHub repository located here:

<https://github.com/spenceraxani/CosmicWatch-Desktop-Muon-Detector-v3C>

The CosmicWatch v3C is a compact, self-contained particle detector designed for measuring ionizing radiation, with a particular focus on cosmic-ray muons. Each detector uses a plastic scintillator coupled to a silicon photomultiplier (SiPM). When a charged particle passes through the scintillator and deposits energy, a small amount of that energy is converted into light. The SiPM detects this light, and the detector electronics amplify, digitize, and record the resulting signal.

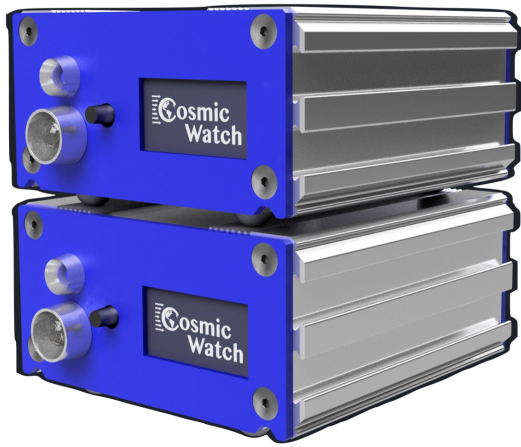


Figure 1: A rendering of a pair of v3C detectors.

The v3C is designed to operate as a complete, standalone detector. It includes an OLED display for viewing detector status and count rates, microSD storage for long-term data collection, USB connectivity for power and real-time data streaming, and sensors for measuring environmental and detector conditions. Recorded event data can include the event time, signal amplitude, coincidence status, detector dead time, temperature, atmospheric pressure, acceleration, and angular velocity.

A single v3C detector can be used to observe individual particle interactions, measure radiation count rates, energy deposition, and investigate changes in the local particle environment. Two v3C detectors can also be connected and operated in coincidence mode. In this configuration, events detected simultaneously by both detectors are identified and recorded, providing a powerful method for selecting penetrating cosmic-ray muons

while suppressing many sources of local radioactive background.

The detector also provides direct access to the SiPM signal, allowing particle-induced waveforms to be viewed using an oscilloscope, giving access to the nanosecond rise times of the signal, as well as capable of identifying single photons. This makes the v3C useful both as a simple counting instrument and as a hands-on particle detector for more advanced laboratory measurements.

This manual covers:

- Detector components and connections

- Initial setup and power-on
- Basic detector operation
- Understanding the indicator lights and detector status
- Recording and accessing data
- Connecting two detectors for coincidence measurements
- Using the analog signal output with an oscilloscope
- Understanding the recorded data fields
- Basic example measurements
- Troubleshooting and recommended operating practices

The CosmicWatch v3C is supplied as a fully assembled and tested detector. No assembly, soldering, programming, or electronics construction is required for normal operation. Users can begin taking measurements by supplying power to the detector and following the Quick Start instructions in this manual.

The CosmicWatch v3C is intended for education, outreach, laboratory instruction, student research, and experimental measurements of cosmic radiation. Its combination of simple standalone operation and access to detailed event-by-event data allows it to be used for everything from introductory demonstrations of cosmic rays to extended quantitative measurements of particle rates and environmental effects.

2 SAFETY AND HANDLING

The CosmicWatch v3C is a low-voltage, low-power particle detector intended for normal laboratory, classroom, outreach, and research use. The detector is supplied fully assembled and tested and does not need to be opened or disassembled during normal operation.

- **Power:** Power the detector using a standard USB power source. This may include a computer USB port, USB power adapter, USB power bank, or cell phone. Use only undamaged USB cables and power sources. Voltage input needs to be 1.8–5.5 V DC.
- **No radioactive source:** The CosmicWatch v3C detects naturally occurring ionizing radiation but does not contain a radioactive source and does not produce ionizing radiation.
- **Keep the detector dry:** The detector and its electronics are not waterproof. Do not expose the detector to rain, liquids, condensation, or excessive humidity.

- **Do not open the detector during normal use:** The detector contains a light-sensitive scintillator and silicon photomultiplier (SiPM) assembly. Opening or modifying the detector may expose the photosensor to ambient light, damage internal components, or affect detector performance.
- **Handle with care:** Avoid dropping the detector or subjecting it to excessive mechanical shock. Although the detector is designed to be portable, impacts may damage the scintillator, SiPM, connectors, or internal electronics.
- **Connections:** Do not force cables into the detector connectors. Verify that the appropriate cable and connector are being used before making a connection.
- **Operating environment:** For normal use, operate the detector in a clean, dry environment. Avoid prolonged exposure to extreme temperatures or direct sunlight.

For normal operation, no user access to the internal electronics is required. If the detector behaves unexpectedly, disconnect power and refer to the troubleshooting section of this manual before attempting to modify or repair the detector.

3 WHAT'S INCLUDED

Each CosmicWatch v3C detector is supplied fully assembled, programmed, and tested. The detector is ready to operate without any soldering, assembly, or firmware installation.

A standard CosmicWatch v3C detector includes:

- **CosmicWatch v3C detector**
- **Inserted microSD card** for recording event-by-event data
- **USB cable** for powering the detector and communicating with a computer

If two detectors were purchased as a pair, a **coincidence cable** (6" CAT6) is also provided. This cable connects the two detectors and allows events passing through both detectors to be identified as coincident events.

3.1 Before You Begin

Before operating the detector, briefly inspect the detector and cables for any visible damage that may have occurred during shipping. No software is required for basic operation. The detector

can be powered from a standard 5 V USB source and immediately begins detecting particles. Data can be viewed directly on the detector's OLED display or recorded to the supplied microSD card.

A computer is only required if you would like to stream data in real time, use the CosmicWatch software, modify detector settings, or access more advanced functionality.

4 DETECTOR OVERVIEW

The CosmicWatch v3C is designed to operate as a self-contained particle detector. The front and rear panels provide access to the detector display, status indicators, data storage, power, coincidence connection, and analog detector signal. Figure 2 identifies the main components of the detector. Each component is described briefly below.

- **OLED Display:** Displays information about the current measurement, including the detector run time, number of recorded events, and average event rate. When two detectors are operating in coincidence mode, coincidence information is also displayed.
- **Event LED (White):** Flashes whenever the detector records a particle event. Under typical indoor conditions, this LED will flash regularly as the detector responds to cosmic rays and naturally occurring background radiation.
- **Coincidence LED (Blue):** Flashes when an event is identified as coincident between two connected detectors. Coincident events are particularly useful for identifying penetrating cosmic-ray muons.
- **Reset Button:** Restarts the detector and begins a new measurement. If a microSD card is installed, resetting the detector closes the previous data file and creates a new file.
- **USB Connection:** Provides 5 V power to the detector. When connected to a computer, the USB connection can also be used to stream event data through the serial interface.
- **microSD Card:** Provides standalone event-by-event data recording. When a properly formatted microSD card is installed, the detector automatically records measurement data to a text file. Note direction.
- **Coincidence Connection:** Allows two CosmicWatch v3C detectors to be connected together. When the detectors recognize one another during startup, they enter coincidence mode and can identify events that pass through both detectors.
- **Analog Signal Output:** Provides access to the detector's analog signal for viewing individual particle interactions with an oscilloscope. This output is useful for examining pulse shapes and signal amplitudes in more advanced laboratory measurements.

Figure 2: Overview of the CosmicWatch v3C detector showing the primary controls, indicators, and connections.

4.1 Inside the Detector

At the heart of the CosmicWatch v3C is a plastic scintillator coupled to a silicon photomultiplier (SiPM). When an ionizing particle deposits energy in the scintillator, a small amount of that energy is converted into light. The SiPM converts this light into an electrical pulse.

The detector electronics amplify and process this pulse, and a Raspberry Pi Pico 2040 microcontroller digitizes the signal and records the event. In addition to the particle signal, the detector can record environmental and motion information including temperature, atmospheric pressure, acceleration, and angular velocity.

Users do not need to access the internal components during normal operation.

5 QUICK START

The CosmicWatch v3C is supplied fully assembled, programmed, and tested. No software or computer is required for basic operation. Follow the steps below to begin detecting particles.

5.1 Operating a Single Detector

1. Insert the microSD card.

If you would like to record data, insert the supplied microSD card into the detector before applying power. The detector can also be operated without a microSD card, but event-by-event data will not be saved locally.

2. Connect USB power.

Connect the detector to a standard 5 V USB power source using the supplied USB cable. A computer USB port, USB power adapter, or USB power bank may be used.

3. Wait for the detector to start.

The detector will automatically perform a series of startup checks. After startup is complete, the OLED display will show the detector status and measurement information.

4. Observe particle events.

The detector immediately begins searching for particle interactions. The white event LED will flash whenever an event exceeds the detector trigger threshold.

Under normal indoor conditions, the detector will record both cosmic-ray particles and naturally occurring background radiation. It is therefore normal for the event LED to flash every second or so.

5. Check the OLED display.

The OLED display provides a convenient way to verify that the detector is operating. During a measurement, the display reports information such as the elapsed run time, total number of recorded events, and average event rate.

6. Collect data.

If a microSD card is installed, event-by-event data are automatically written to a text file. No additional action is required to begin recording.

5.2 Ending a Measurement

To end a measurement, disconnect the USB power from the detector. If a microSD card is installed, allow the detector to operate normally for a few seconds before removing power.

Each time the detector is powered on or reset, a new data file is created on the microSD card. This makes it easy to separate measurements into individual runs.

5.3 Your First Measurement

For a simple first measurement, place the detector flat on a table and allow it to operate for several minutes. You should see the event LED flash as particles are detected and the event count on the OLED display increase.

The detector is now taking data.

More detailed instructions for recording data, interpreting the measurements, and operating two detectors in coincidence are provided in the following sections.

6 USING TWO DETECTORS IN COINCIDENCE MODE

Two CosmicWatch v3C detectors can be connected together to operate in *coincidence mode*. Coincidence mode identifies particle events that are detected by both detectors at approximately the same time.

This is particularly useful for identifying cosmic-ray muons. A penetrating muon can pass through both detectors and produce a signal in each, while many sources of local radioactive background will trigger only one detector.

6.1 Connecting Two Detectors

1. Place the two detectors near one another. For a simple cosmic-ray muon measurement, stack one detector directly above the other.
2. Connect the detectors using the supplied coincidence cable.
3. Connect USB power to **one** of the detectors. The second detector receives power through the coincidence cable and does not require a separate USB power connection.
4. Power on or reset the detectors while they are connected.
5. During startup, the detectors automatically check for the presence of another detector. If the connection is recognized, the detectors will indicate that they have entered coincidence mode.

6.2 Taking Coincidence Data

Once coincidence mode is active, both detectors continue to record their normal particle events.

When a detector triggers, it communicates the event to the connected detector. If both detectors observe an event within the coincidence time window, the event is identified as a *coincident event*.

Coincident events cause the blue coincidence LED to flash and are recorded with a coincidence flag in the saved event data.

All events are still recorded in coincidence mode.

Coincidence mode does not discard single-detector events. Instead, events identified as coincident are marked in the recorded data so that they can be selected during analysis.

6.3 A Simple Muon Telescope

For a simple cosmic-ray measurement, place the two detectors horizontally with one directly above the other. A particle traveling through both scintillators can then produce a coincident event.

The relative positions of the detectors determine which particle trajectories can produce a coincidence. Moving the detectors farther apart reduces the range of accepted particle directions, while changing their relative orientation allows the detector pair to probe different directions.

In this configuration, the two CosmicWatch detectors form a simple *muon telescope*.

Coincidence measurements strongly suppress many backgrounds that trigger only one detector and provide a substantially cleaner sample of penetrating cosmic-ray muons than a single detector operating alone.

7 RECORDING DATA

The CosmicWatch v3C provides several ways to monitor and record measurements. For most users, recording data directly to the microSD card is the simplest and recommended method.

7.1 OLED Display

The OLED display provides a quick way to monitor the detector without connecting it to a computer. During operation, the display reports information including the total run time, number of detected events, and average event rate.

When operating in coincidence mode, the display also reports the number and rate of coincident events.

The OLED display is useful for demonstrations and measurements where only the total number of events or average event rate is required.

7.2 Recording to a microSD Card

For most measurements, the recommended method is to record data directly to a microSD card.

Insert the microSD card before powering on or resetting the detector. During startup, the detector will automatically detect the card and begin recording data.

Each time the detector is powered on or reset, a **new data file** is created. Measurements from different runs are therefore automatically stored in separate files.

Data are stored as plain-text (`.txt`) files containing one row for each recorded event. These files

can be opened using a text editor or imported into common analysis programs such as Python, MATLAB, Excel, or similar software.

microSD Card Requirements

The microSD card must be formatted using the **exFAT** filesystem.

A relatively small microSD card provides sufficient storage for long-duration measurements because each particle event requires only a small amount of data. We recommend using a 4 GB microSD card. This allows for a few months of data to be stored.

Data File Names

The detector automatically assigns a name to each data file. The filename contains the detector name, operating mode, and a sequential file number.

For example: **AxTech_C_002.txt**

In this example, AxTech is the detector name, C indicates that the detector was operating in coincidence mode, and 002 indicates the run number.

Files recorded while operating as a single detector use M to indicate *master mode*.

7.3 Recording Data with a Computer

When the CosmicWatch v3C is connected to a computer through USB, event data are also transmitted through the detector's serial connection.

The serial connection operates at: 115200 baud.

The serial output can be viewed using a serial monitor or recorded directly using the CosmicWatch data-acquisition software.

Recording through a computer has the additional advantage that the computer can associate detector events with its global date and time from the computer itself. This is useful for measurements involving multiple detectors, comparisons between different locations, or measurements that require an absolute timestamp.

7.4 Removing the microSD Card

Before removing the microSD card, disconnect power from the detector. Do not remove the card while the detector is actively recording data.

After the detector has been powered off, the microSD card can be removed and inserted into a computer to access the recorded `.txt` files.

The format and meaning of each recorded data column are described in the following section.

8 UNDERSTANDING THE DATA

The CosmicWatch v3C records data on an event-by-event basis. Each time the detector triggers, a new row is added to the data file containing information about the particle interaction and the state of the detector.

Data files are stored as tab-delimited plain-text (`.txt`) files. Lines beginning with the `#` symbol contain comments, configuration information, or detector diagnostics and are not particle events.

The event data contain the following columns:

8.1 Coincidence Flag

The coincidence flag is particularly useful when two detectors are connected together. An event with `Flag = 0`, was recorded by the detector but was not identified as coincident with the connected detector.

An event with `Flag = 1` was detected in coincidence with the connected detector. Selecting events with `Flag = 1` provides a convenient way to obtain a substantially cleaner sample of penetrating cosmic-ray muons.

8.2 Calculating the Event Rate

For short measurements, the approximate average event rate can be calculated by dividing the number of recorded events by the elapsed run time.

For more accurate measurements, the detector dead time should be removed from the total run time. The detector *live time* is

Column	Description
Event	Sequential event number. The counter increases by one each time the detector records an event.
Timestamp [s]	Elapsed time since the detector began the current run, measured in seconds. This is a relative detector timestamp rather than the absolute date and time.
Flag	Coincidence flag. A value of 1 indicates that the event was identified as coincident with an event in the connected detector. A value of 0 indicates a non-coincident event.
ADC [12b]	Peak signal amplitude measured by the detector's 12-bit analog-to-digital converter (ADC). The ADC has a range from 0 to 4095.
SiPM [mV]	Calculated pulse height of the SiPM signal in millivolts. The pulse height is related to the amount of scintillation light detected and therefore to the energy deposited in the scintillator.
Dead Time [s]	Cumulative amount of time for which the detector has been unable to search for new events since the beginning of the run.
Temp. [°C]	Temperature measured by the detector's environmental sensor.
Press. [Pa]	Atmospheric pressure measured by the detector's environmental sensor, reported in pascals.
Accel. (X:Y) [g]	Acceleration measured along the detector's three axes, expressed in units of gravitational acceleration, g .
Gyro. (X:Y) [deg/s]	Angular velocity measured about the detector's three axes, expressed in degrees per second.

Table 1: Event-by-event information recorded by the CosmicWatch v3C.

$$T_{\text{live}} = T_{\text{run}} - T_{\text{dead}}, \quad (1)$$

where T_{run} is the elapsed run time and T_{dead} is the cumulative detector dead time.

The average event rate is then

$$R = \frac{N}{T_{\text{live}}}, \quad (2)$$

where N is the number of recorded events.

For example, if a detector records 900 events during a 300-second measurement and accumulates 2 seconds of dead time,

$$T_{\text{live}} = 300 \text{ s} - 2 \text{ s} = 298 \text{ s}, \quad (3)$$

and the measured event rate is

$$R = \frac{900}{298 \text{ s}} \approx 3.02 \text{ Hz.} \quad (4)$$

8.3 Absolute Date and Time

The timestamp recorded directly by the detector represents the elapsed time since the beginning of the run and should not be interpreted as an absolute global timestamp.

When data are recorded through the CosmicWatch computer software (`import_data.py`), the computer can add additional information to each event, including the detector name, global time, and date from NTP. This is useful when comparing measurements between multiple detectors or measurements made at different locations.

8.4 Detector Diagnostics

When the CosmicWatch v3C starts, it performs a series of diagnostic measurements and checks. Diagnostic information is written as commented lines beginning with the `#` symbol.

These lines provide information about the detector configuration and startup conditions and should generally be retained with the data file. They can be ignored when selecting event rows for analysis.

9 CONFIGURING THE DETECTOR

The CosmicWatch v3C can be configured using the `configure.txt` file stored on the microSD card. This allows common detector settings to be changed without modifying or recompiling the detector firmware.

9.1 Changing the Configuration

To change the detector configuration:

1. Disconnect power from the detector.

2. Remove the microSD card and insert it into a computer.
3. Open the `configure.txt` file using a plain-text editor.
4. Modify the desired settings and save the file.
5. Safely eject the microSD card from the computer and reinstall it in the detector.
6. Power on or reset the detector. The new configuration will be read during startup.

9.2 Detector Name

Each detector can be assigned a unique name. The detector name is useful when operating several CosmicWatch detectors or when collecting data at multiple locations.

The detector name is included in the filenames generated by the detector, making it easier to identify which detector produced each data file.

Use a short, descriptive name without special characters. For example: *Detector01*, or *Hess*.

9.3 OLED Display

The OLED display can be enabled or disabled through the configuration file.

Disabling the OLED may be useful for long-duration measurements where the display is not required. The firmware does move the display by 3 pixels every minute to minimize OLED burnin.

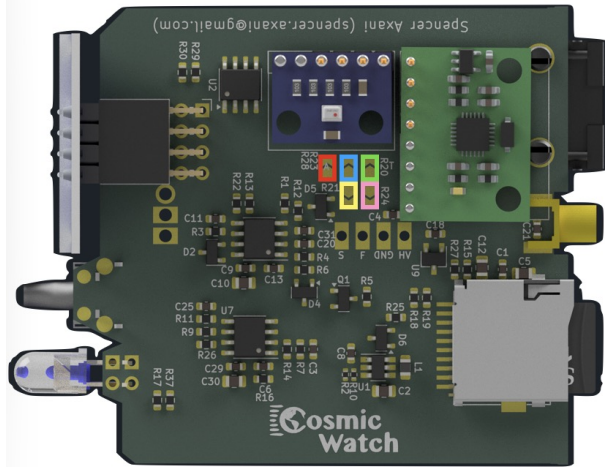
Changing this setting does not disable particle detection or data recording.

9.4 Restoring the Configuration

If the detector does not operate as expected after changing the configuration, simply delete the configure file, and the detector will make a new one during bootup.

When editing the configuration file, change only the values associated with the desired settings. Do not change parameter names or the overall formatting of the file unless instructed to do so.

If you would like to output a different part of the circuit to the MCX analog connector, you can short the appropriate connection show in Fig. 3. For example, if you would like to look at the



Jumper	Purpose
 SiPM (Slow)	Raw SiPM pulse
 SiPM (Fast)	Fast SiPM Pulse, with sub ns rise time.
 First Stage Amp	Amplified stage of the raw SiPM pulse.
 Second Stage Amp	Second amplification stage
 Trigger Signal	Signal from comparator circuit

Figure 3: Sending different parts of the analog circuit to the MCX connector. Only short one at a time. Default is SiPM (Slow).

fast signals from the SiPM (MicroFC 60035 Fast output), short the connection highlighted in pink, and remove the SiPM (Slow) short.

10 UPDATING THE FIRMWARE

The CosmicWatch v3C firmware runs on the detector's Raspberry Pi Pico 2040 microcontroller. Firmware updates may occasionally be released to add features, improve detector performance, or correct software issues.

Firmware is distributed as a .uf2 file via the github link on the first page and can be installed using a computer and a USB cable. No programming software is required.

10.1 Before Updating

Before beginning, obtain the correct CosmicWatch v3C firmware file. The firmware file will have a name ending in **.uf2**.

Disconnect the detector from USB power before beginning the update.

10.2 Installing New Firmware

1. **Disconnect the detector.** Disconnect the USB cable from the CosmicWatch.

2. **Hold the BOOTSEL button.**

Locate and press the **BOOTSEL** button on the Raspberry Pi Pico inside the detector. This requires you to remove the four M2 screws located on the back of the detector.

3. **Connect the USB cable.**

While continuing to hold the BOOTSEL button, connect the detector to your computer using the USB cable.

4. **Release the BOOTSEL button.**

The Raspberry Pi Pico should appear on the computer as a removable USB drive named **RPI-RP2**.

5. **Copy the firmware file.**

Copy the CosmicWatch v3C .uf2 firmware file onto the RPI-RP2 drive.

6. **Wait for the update to complete.**

After the file has been copied, the RPI-RP2 drive will automatically disappear and the Raspberry Pi Pico will restart using the new firmware.

7. **Verify detector operation.**

Allow the detector to complete its normal startup sequence. Verify that the OLED display, event LED, and other detector functions operate normally.

10.3 If the Update Does Not Work

If the RPI-RP2 drive does not appear:

- disconnect the USB cable;
- press and hold the BOOTSEL button;
- reconnect the USB cable while continuing to hold BOOTSEL; and
- release BOOTSEL only after the USB connection has been made.

If the detector does not operate normally after installing new firmware, repeat the procedure and confirm that the correct CosmicWatch v3C .uf2 file was used.

10.4 Configuration and Data

Updating the firmware does not require reformatting the microSD card. Existing measurement files stored on the microSD card should be retained.

After updating the firmware, verify the detector configuration and confirm that the detector starts normally before beginning a new measurement.

11 OSCILLOSCOPE AND ANALOG OUTPUT

The CosmicWatch v3C provides a direct analog signal (SiPM) output that allows the detector pulses to be viewed using an oscilloscope. This is useful for demonstrations, detector diagnostics, and experiments involving pulse shape or pulse amplitude.

11.1 Connecting an Oscilloscope

Connect the detector's analog output (MCX) to an oscilloscope using an appropriate coaxial cable. I suggest using an MCX to BNC cable.

The detector should be powered and operating normally while using the analog output. Particle interactions will then appear as individual voltage pulses on the oscilloscope.

The analog output can be monitored while the detector simultaneously records events to the microSD card.

11.2 Viewing Particle Pulses

When a charged particle passes through the plastic scintillator, it produces a short burst of scintillation light. The SiPM converts this light into an electrical pulse. The analog output allows this pulse to be observed directly.

To initially locate the pulses:

1. Connect the analog output to an oscilloscope.
2. Set the oscilloscope input to DC coupling.
3. Begin with a time scale of approximately 100 nanoseconds per division.

4. Adjust the vertical scale to 10 mV per division.
5. Set the oscilloscope trigger level above the baseline noise and adjust it until pulses trigger reliably.

Once a pulse is visible, the horizontal and vertical scales can be adjusted to examine its shape in greater detail.

11.3 Pulse Amplitude

The amplitude of a detector pulse depends on the amount of scintillation light detected by the SiPM. In general, larger energy depositions produce more scintillation light and therefore larger electrical signals.

The analog output can therefore be used to compare the relative sizes of different particle interactions.

Small pulses are also visible every microsecond or so. These are signals from either single photons or thermal photoelectron emission.

11.4 Typical Applications

The analog output can be used to:

- observe individual SiPM pulses;
- demonstrate how particle interactions produce electrical signals;
- study pulse amplitudes and pulse shapes;
- investigate detector noise and trigger thresholds;
- compare signals from different particle interactions; and
- connect the CosmicWatch to external data-acquisition or laboratory electronics.

Using an oscilloscope is optional and is not required for normal CosmicWatch operation or data collection.

12 EXAMPLE MEASUREMENTS

The CosmicWatch v3C can be used for a variety of cosmic-ray and environmental measurements. This section shows several examples of measurements that can be performed directly using the quantities recorded by the detector.

The figures below show example data collected with two CosmicWatch v3C detectors operating in coincidence mode.

12.1 Measure the Cosmic-Ray Rate

One of the simplest measurements is to determine the particle detection rate as a function of time.

The average event rate is calculated from the number of recorded events and the detector live time,

$$R = \frac{N}{T_{\text{live}}}. \quad (5)$$

When two detectors are operated in coincidence mode, the rates of all events, non-coincident events, and coincident events can be examined separately.

Figure 4 shows an example measurement lasting approximately 14 hours. In this measurement, the average rates were approximately 1.39 Hz for all events, 1.18 Hz for non-coincident events, and 0.205 Hz for coincident events.

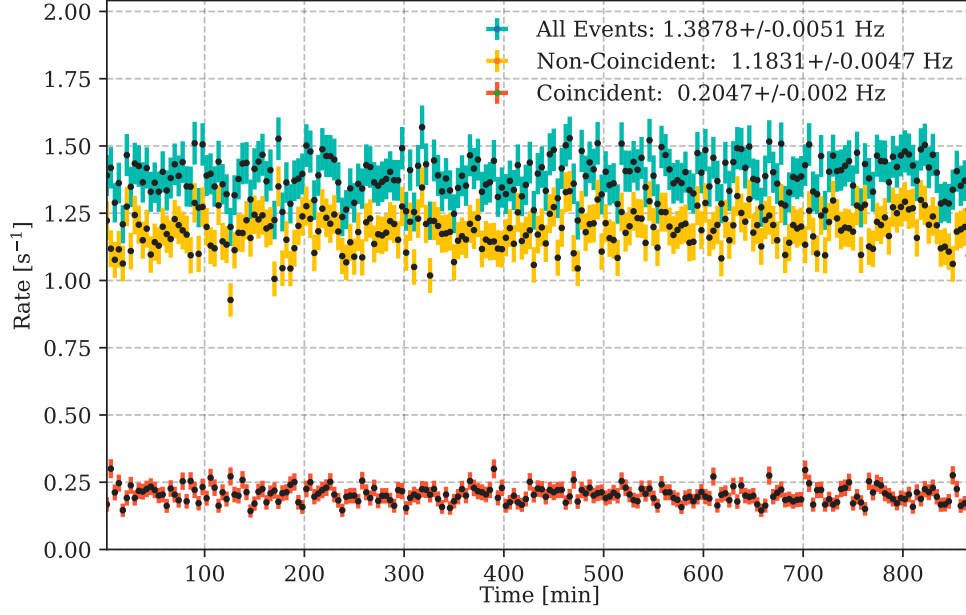


Figure 4: Example event rate as a function of time. The total event rate is separated into non-coincident and coincident events. Statistical uncertainties are shown for each time bin.

The measured rate fluctuates from one time interval to another because particle detection is a statistical process. Longer measurement intervals contain more events and therefore provide a more precise estimate of the average rate.

The distribution of measured rates over the complete run can also be examined, as shown in Figure 5.

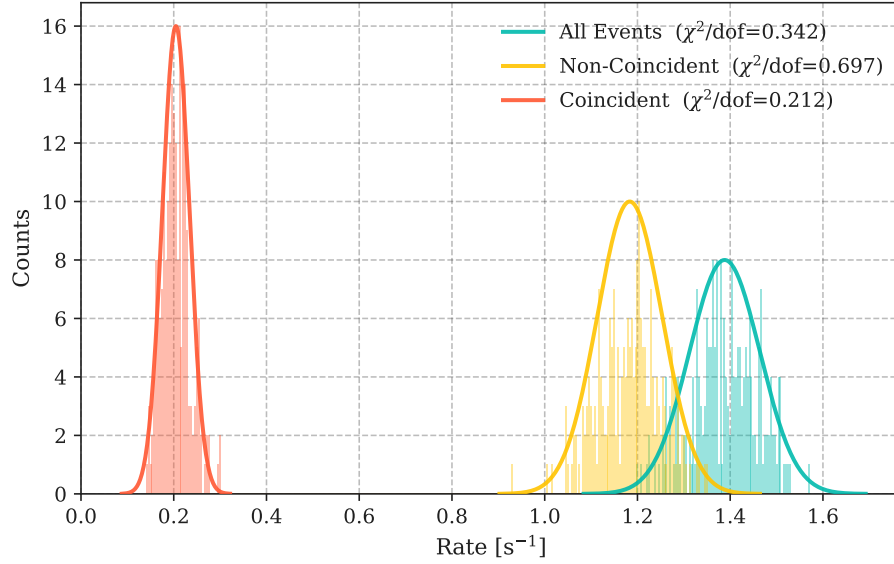


Figure 5: Distribution of the measured event rates during an example run. The total, non-coincident, and coincident event samples are shown separately.

12.2 Detect Cosmic-Ray Muons

Two detectors can be used together to obtain a cleaner sample of cosmic-ray muons. Connect the detectors in coincidence mode and place one detector directly above the other.

Events recorded with

$$\text{Flag} = 1$$

were detected by both detectors within the coincidence time window. These events are much more likely to have been produced by penetrating cosmic-ray particles passing through both scintillators.

The coincidence rate can be measured in exactly the same way as the total event rate. In Figure 4, the coincident sample has an average rate of approximately 0.205,Hz, corresponding to roughly one coincident event every five seconds.

Changing the separation or relative orientation of the two detectors changes the range of particle trajectories capable of passing through both scintillators. This allows the detector pair to be used as a simple directional muon telescope.

12.3 Coincidence Geometry

Two CosmicWatch detectors operating in coincidence form a simple directional particle telescope. A coincident event occurs when a particle passes through the scintillator in **both** detectors. The relative separation and orientation of the detectors therefore determine the range of particle trajectories that can be measured.

Figure 6 illustrates several possible detector configurations.

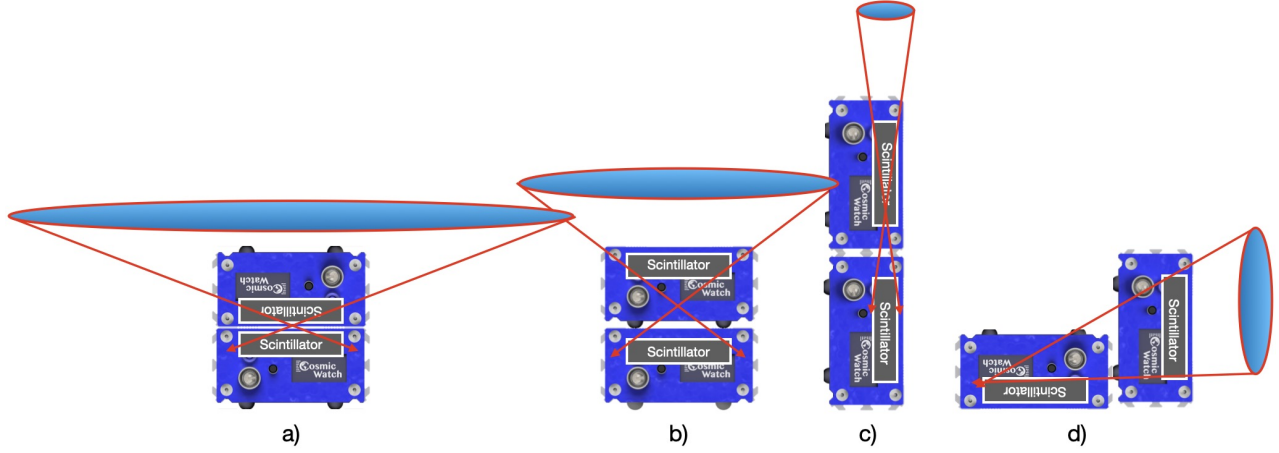


Figure 6: Example coincidence configurations for two CosmicWatch v3C detectors. The blue regions illustrate the approximate angular acceptance of each configuration. (a) Two closely spaced detectors provide a wide angular acceptance. (b) A modest separation reduces the angular acceptance. (c) A larger separation produces a narrow acceptance around the vertical direction. (d) Reorienting the detector pair allows particles traveling in other directions to be selected.

Detector Separation. When the detectors are placed close together, as shown in Figure 6(a), particles arriving over a relatively wide range of angles can pass through both scintillators. Moving the detectors farther apart, as shown in (b) and (c), progressively narrows the range of accepted trajectories.

Increasing the separation therefore improves the directional selectivity of the detector pair, but also reduces the coincidence rate because fewer particle trajectories intersect both scintillators.

Detector Orientation. The direction selected by the telescope can be changed by rotating the detector pair. In Figure 6(c), the detectors are arranged to preferentially select particles traveling approximately vertically. In Figure 6(d), the detectors are arranged to select particles traveling closer to the horizontal direction.

By repeating the same measurement at different orientations, the CosmicWatch can be used to investigate the angular dependence of the cosmic-ray muon flux.

For quantitative comparisons, keep the detector separation and measurement duration the same at each orientation. This ensures that changes in the measured coincidence rate primarily reflect changes in the incident particle flux rather than changes in the detector geometry.

12.4 Pulse-Height Distribution

The amplitude of each detector signal is recorded by the ADC. Examining the distribution of ADC values provides information about the relative sizes of the detected particle signals.

Figure 7 shows an example pulse-height distribution for all events, non-coincident events, and coincident events.

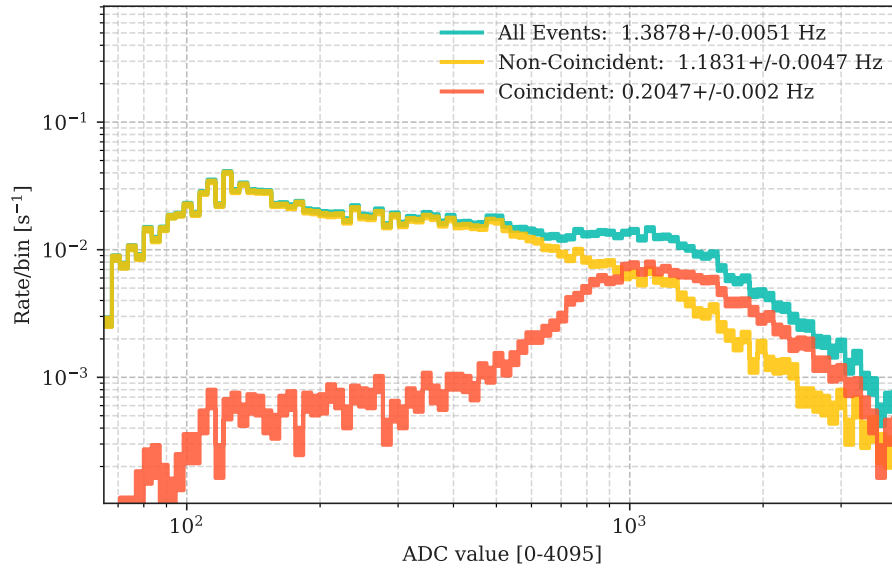


Figure 7: Example ADC pulse-height distributions for all events, non-coincident events, and coincident events. Coincident events preferentially populate a different portion of the pulse-height distribution than the complete event sample.

Comparing these distributions is a useful way to study the detector response to different populations of events. Coincidence selection provides a cleaner sample of penetrating cosmic-ray particles and can therefore produce a noticeably different pulse-height distribution from the full single-detector sample.

12.5 Time Between Events

Another useful quantity is the time between consecutive detector triggers,

$$\Delta t = t_{i+1} - t_i. \quad (6)$$

A histogram of Δt provides a simple way to study the statistical timing of particle arrivals.

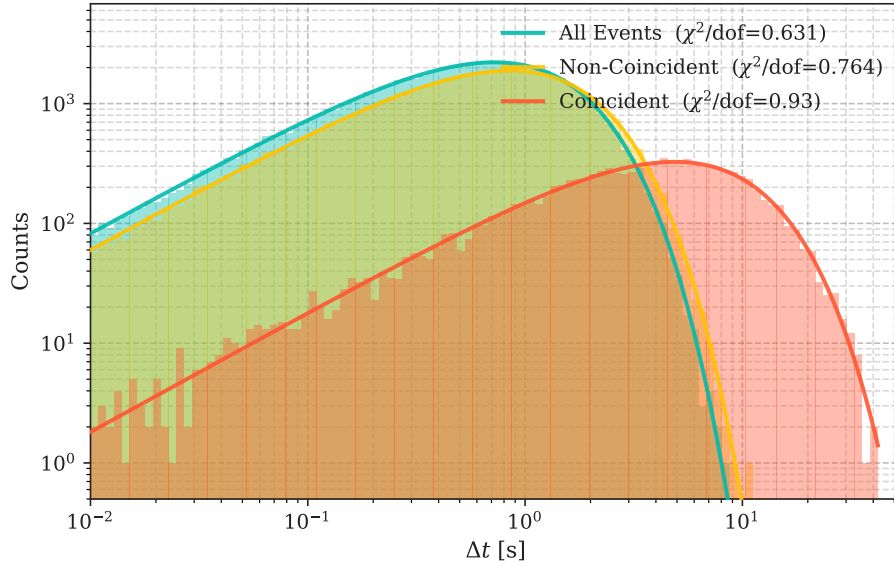


Figure 8: Distribution of the time between consecutive events for all events, non-coincident events, and coincident events.

Because coincident events occur less frequently than single-detector triggers, their characteristic time between events is longer.

12.6 Detector Dead Time

After recording an event, the detector requires a short period to process and store the measurement. During this interval, the detector is not available to record another event. This is referred to as the detector *dead time*.

The dead-time fraction can be monitored throughout a measurement,

$$f_{\text{dead}} = \frac{T_{\text{dead}}}{T_{\text{measurement}}}. \quad (7)$$

Figure 9 shows the dead-time fraction measured throughout an example run.

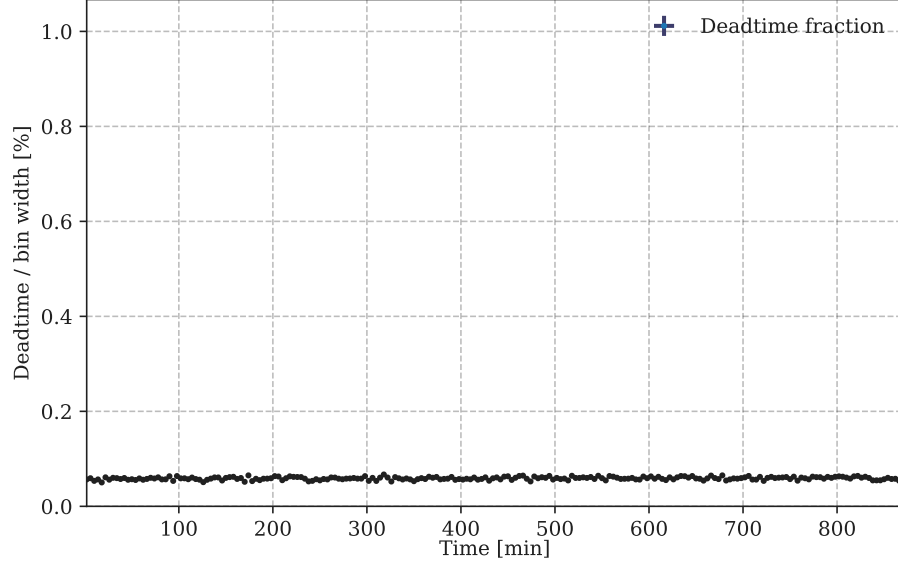


Figure 9: Example detector dead-time fraction as a function of time. In this measurement, the dead-time fraction remains small and approximately constant throughout the run.

Monitoring the dead time is particularly useful when comparing measurements made under conditions with substantially different event rates.

12.7 Environmental Conditions

The CosmicWatch v3C records atmospheric pressure and temperature together with the particle data. These measurements are useful for monitoring environmental conditions during long-duration measurements and investigating possible correlations with the measured cosmic-ray rate.

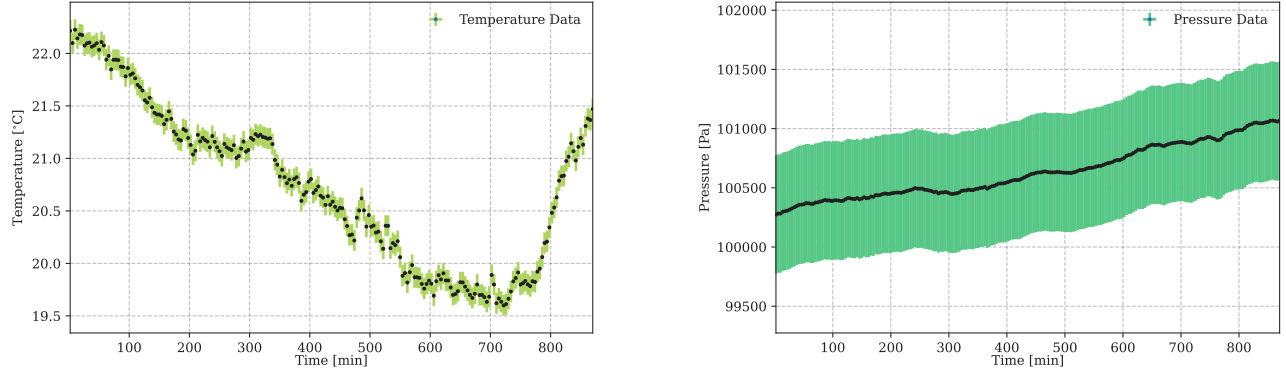


Figure 10: Temperature (left) and atmospheric pressure (right) recorded by the CosmicWatch v3C during an example measurement.

Recording these quantities alongside the particle data is particularly useful for long-duration measurements. For example, changes in atmospheric pressure can be compared with changes in the measured cosmic-ray rate, while temperature measurements can be used to monitor the environmental conditions of the detector.

12.8 Detector Orientation and Motion

The CosmicWatch v3C records three-axis acceleration and angular velocity. These measurements can be used to determine the orientation of the detector and identify periods during which it was moved or rotated.

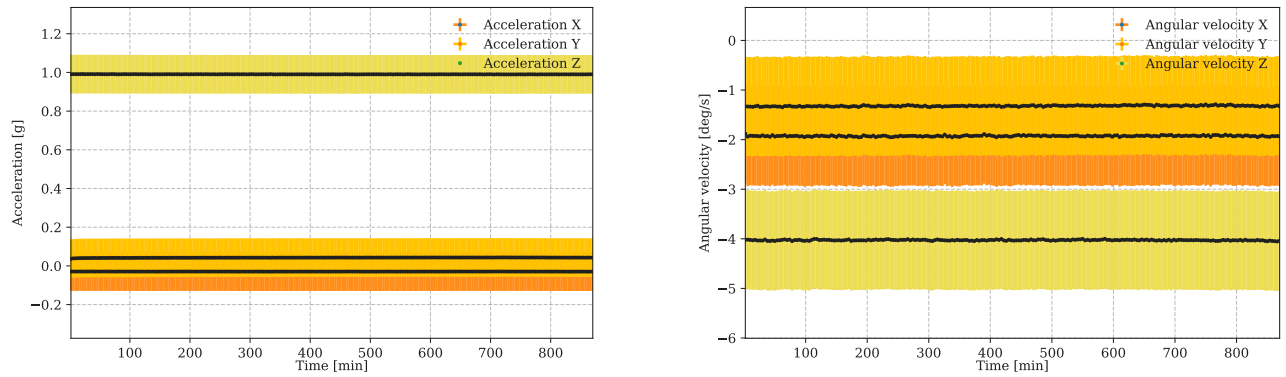


Figure 11: Three-axis acceleration (left) and angular velocity (right) recorded by the CosmicWatch v3C.

For a stationary detector, the accelerometer measures the direction of gravity, allowing the detector orientation to be determined. The gyroscope can be used to identify periods when the

detector is being rotated. These measurements are particularly useful for portable experiments, balloon measurements, vehicle-based measurements, or experiments in which the detector orientation is intentionally changed.

12.9 Additional Experiments

The quantities recorded by the CosmicWatch v3C can be combined to perform more advanced measurements. Examples include:

- measuring the cosmic-ray rate over many hours or days;
- comparing cosmic-ray rates at different altitudes or locations;
- measuring the angular dependence of the cosmic-ray muon flux using two detectors in coincidence;
- studying the relationship between atmospheric pressure and cosmic-ray rate;
- comparing the pulse-height distributions of coincident and non-coincident events;
- investigating the statistical distribution of particle arrival times; and
- using the accelerometer and gyroscope to associate particle measurements with detector position or motion.

Because the particle, environmental, and motion measurements are recorded together, a single CosmicWatch data file can be used to investigate several of these effects simultaneously.

13 TROUBLESHOOTING

Most problems with the CosmicWatch v3C can be identified using the OLED display, status LEDs, and startup diagnostic messages. Common problems and suggested solutions are listed below.

13.1 Restarting the Detector

Many temporary problems can be resolved by restarting the detector. Press the reset button or disconnect and reconnect USB power.

Problem	Suggested Solution
Detector does not turn on	Check the USB cable and power source. Try a different USB cable or standard 5,V USB power source.
OLED does not turn on	Confirm that the detector is receiving power. The OLED may also have been disabled in the <code>configure.txt</code> file.
No events are being recorded	Allow the detector to operate for several minutes and watch the event LED. If no events are observed, reset the detector and check the startup diagnostic information for errors.
microSD card is not detected	Turn off the detector and reinstall the microSD card. Verify that the card is formatted using the exFAT filesystem. Then restart the detector and check the startup messages.
Data are not being saved	Confirm that the microSD card was installed before the detector was powered on or reset. Check that sufficient free space is available on the card and that the card can be read by a computer.
Two detectors do not enter coincidence mode	Confirm that the coincidence cable is fully connected to both detectors. Power on or reset both detectors while they are connected.
No coincidence events are observed	Confirm that both detectors are operating in coincidence mode. Stack the detectors directly on top of one another and allow them to operate for several minutes.
Event rate appears unusually high	Check that the detector is not located near a radioactive source or other source of ionizing radiation. Reset the detector and verify that the high rate persists.
Event rate appears unusually low	Allow the detector to operate for a longer period to distinguish a true rate change from normal statistical fluctuations. Check the detector startup diagnostics and verify that events are being recorded.
Computer does not detect the detector	Try a different USB cable or USB port. Some USB cables provide power but do not contain data connections. Verify that the detector appears as a serial device on the computer.
Serial data are unreadable	Verify that the serial connection is configured for 115200 baud .
Detector behaves unexpectedly after changing settings	Restore the original <code>configure.txt</code> by deleting the file, and allow the detector to make a new <code>configure.txt</code> file.

Table 2: Common CosmicWatch v3C troubleshooting steps.

A new measurement run begins after the detector restarts. If a microSD card is installed, a new data file will be created.

13.2 Startup Diagnostics

During startup, the CosmicWatch v3C performs several checks of the detector hardware and connected peripherals. Diagnostic information is also transmitted through the USB serial connection and written to the data file as commented lines when a microSD card is installed.

If a detector is not operating normally, these diagnostic messages can help identify the source of the problem.

13.3 Still Having Problems?

If the problem persists, record the following information before requesting support:

- a description of the problem;
- the startup diagnostic messages;
- a copy of a recent data file, if available; and
- photographs of the detector and connections if the problem appears to be hardware-related.

Feel free to reach out to saxani@axtechscientific.org for questions.

14 SPECIFICATIONS

The principal specifications of the CosmicWatch v3C are summarized below.

14.1 Power

For normal operation, power the CosmicWatch v3C using a standard 5,V USB source. Suitable sources include a computer USB port, USB wall adapter, or USB power bank. It will draw approximately 0.5 W per detector.

Parameter	Specification
Detector type	Plastic scintillator with silicon photomultiplier (SiPM) read-out
Scintillator dimensions	$3.9 \times 3.9 \times 1.0, \text{cm}^3$
Primary application	Detection of ionizing radiation and cosmic-ray muons
Microcontroller	Raspberry Pi Pico
ADC resolution	12-bit
Local data storage	microSD card
microSD filesystem	exFAT
Data format	Tab-delimited plain-text (<code>.txt</code>) files
USB power	Standard 5,V USB
USB communication	Serial communication at 115200 baud
Display	0.96-inch OLED display
Coincidence operation	Two v3C detectors can be connected for coincidence measurements
Coincidence data	Coincident events identified in recorded event data
Analog output	Direct analog detector signal for oscilloscope or external electronics
Environmental measurements	Temperature and atmospheric pressure
Motion measurements	Three-axis acceleration and three-axis angular velocity
Recorded event quantities	Timestamp, coincidence flag, ADC amplitude, SiPM pulse height, cumulative dead time, temperature, pressure, acceleration, and angular velocity

Table 3: Principal specifications of the CosmicWatch v3C detector.

When two detectors are connected in coincidence mode, only one detector requires a USB power connection. Power is supplied to the second detector through the coincidence connection.

14.2 Detector Medium

The CosmicWatch v3C uses a plastic scintillator coupled to a silicon photomultiplier (SiPM). The active scintillator volume is

$$3.9 \times 3.9 \times 1.0 \text{ cm}^3. \quad (8)$$

Ionizing particles passing through the scintillator produce scintillation light. The SiPM detects this light and converts it into an electrical signal that is processed by the detector electronics.

14.3 Data Acquisition

Each triggered event is digitized and recorded by the onboard electronics. Measurements can be stored directly on the microSD card or transmitted to a computer through the USB serial connection.

The detector can operate completely independently of a computer when recording data to the microSD card.

15 SUPPORT AND ADDITIONAL RESOURCES

Additional CosmicWatch documentation, software, firmware, and example analysis tools are available through GitHub repository.

<https://github.com/spenceraxani/CosmicWatch-Desktop-Muon-Detector-v3C>

These resources include:

- detector software and firmware updates;
- example Python analysis scripts;

- example measurements and educational activities; and
- additional technical documentation.

15.1 Technical Support

If you experience a problem with your CosmicWatch v3C detector, please include a description of the problem, and any relevant startup diagnostic messages when requesting support. My email address is saxani@axtechscientific.org.

If possible, also include a recent data file and photographs of the detector setup or connections.

15.2 CosmicWatch Project

For additional information about the CosmicWatch project, detector documentation, and software, please visit the CosmicWatch website and GitHub repository.

CosmicWatch: The Desktop Muon Detector